

Lecture #8

PREDICATE LOGIC

Predicates, Quantifiers, Free and Bound Variables,
Formalization of Statements

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Propositional logic cannot adequately express the meaning of all statements in mathematics and in natural language.

Now we will introduce a more powerful type of logic called **predicate logic**. We will see how predicate logic can be used to express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects.

To understand predicate logic, we first need to introduce the concept of a predicate.

Statements involving variables, such as

$x > 3,$ $x = y + 3,$ $x + y = z,$

and

“computer x is under attack by an intruder,”

and

“computer x is functioning properly,”

are often found in mathematical assertions, in computer programs, and in system specifications.

Note: These statements are neither true nor false when the values of the variables are not specified.

The statement “ x is greater than 3” has 2 parts.

- *The first part*, the variable x , is the subject of the statement.
- *The second part* – the **predicate**, “is greater than 3” – refers to a property that the subject of the statement can have.

We can denote the statement “ x is greater than 3” by $P(x)$, where P denotes the predicate “is greater than 3” and x is the variable.

The statement $P(x)$ is also said to be the value of the **propositional function** P at x . Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.

Example. Let $Q(x, y)$ denote the statement “ $x = y + 3$.” What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Solution:

To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$.

Hence, $Q(1, 2)$ is the statement “ $1 = 2 + 3$,” which is false.

The statement $Q(3, 0)$ is the proposition “ $3 = 0 + 3$,” which is true.

Definition. A *predicate* is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables. The *domain* of a predicate variable is the set of all values that may be substituted in place of the variable.

Definition. If $P(x)$ is a predicate and x has domain D , the *truth set* of $P(x)$ is the set of all elements of D that make $P(x)$ true when they are substituted for x . The truth set of $P(x)$ is denoted

$$\{ x \in D \mid P(x) \}$$

Quantifiers

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value.

However, there is another important way, called **quantification**, to create a proposition from a propositional function. Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words *all*, *some*, *many*, *none*, and *few* are used in quantifications. The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

Definition. The universal quantifier Let $Q(x)$ be a predicate and D the domain of x . A ***universal statement*** is a statement of the form

$$\forall x \in D [Q(x)]$$

- It is defined to be true if, and only if, $Q(x)$ is true for each individual $x \in D$.
- It is defined to be false if, and only if, $Q(x)$ is false for at least one $x \in D$.

A value for x for which $Q(x)$ is false is called a ***counterexample*** to the universal statement.

Remark. If the context clearly a domain set, then usually write $\forall x$ instead of $\forall x \in D$.

Truth and Falsity of Universal Statements

Example. Let $D = \{1,2,3,4,5\}$, and consider the statement
$$\forall x \in D [x^2 \geq x]$$

Write one way to read this statement out loud and show that it is true.

Example. Consider the statement

$$\forall x \in R [x^2 \geq x]$$

Find a counterexample to show that this statement is false.

Universal quantification: example

Example. Let $P(x)$ be the statement “ $x + 1 > x$.” What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Solution: Because $P(x)$ is true for all real numbers x , the quantification $\forall x P(x)$ is true.

Remark

1) Generally, an implicit assumption is made that all domains of discourse for quantifiers are nonempty. Note that if the domain is empty, then $\forall x P(x)$ is true for any propositional function $P(x)$ because there are no elements x in the domain for which $P(x)$ is false.

2) Besides “for all” and “for every,” universal quantification can be expressed in many other ways, including “all of,” “for each,” “given any,” “for arbitrary,” “for each,” and “for any.”

Example. Let $Q(x)$ be the statement “ $x < 2$.” What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Solution:

$Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false.

That is, $x = 3$ is a counterexample for the statement $\forall x Q(x)$.

Thus $\forall x Q(x)$ is false.

The existential quantifier

Definition. Let $Q(x)$ be a predicate and D the domain of x . An *existential statement* is a statement of the form “ $\exists x \in D [Q(x)]$ ”.

- It is defined to be true if, and only if, $Q(x)$ is true for at least one $x \in D$.
- It is false if, and only if, $Q(x)$ is false for all $x \in D$.

If the context clearly a domain set, then usually write $\exists x$ instead of $\exists x \in D$.

Example. Consider the statement

$$\exists m \in \mathbb{Z}^+ [m^2 = m]$$

Write one way to read this statement out loud and show that it is true.

Example. Let $D = \{5,6,7,8\}$ and consider the statement

$$\exists m \in D [m^2 = m]$$

Show that this statement is false.

Besides the phrase “there exists,” we can also express existential quantification in many other ways, such as by using the words “for some,” “for at least one,” or “there is.”

The existential quantification $\exists x P(x)$ is read as

“There is an x such that $P(x)$,”

“There is at least one x such that $P(x)$,”

or

“For some x $P(x)$.”

Example. Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Solution: Because “ $x > 3$ ” is sometimes true –

for instance, when $x = 4$

the existential quantification of $P(x)$, which is $\exists x P(x)$, is true.

Remark

- 1) Generally, an implicit assumption is made that all domains of discourse for quantifiers are nonempty. If the domain is empty, then $\exists x Q(x)$ is false whenever $Q(x)$ is a propositional function because when the domain is empty, there can be no element x in the domain for which $Q(x)$ is true.
- 2) When all elements in the domain can be listed – say, $x_1, x_2, \dots, x_n$ – the existential quantification $\exists x P(x)$ is the same as the disjunction

$$P(x_1) \vee P(x_2) \vee \dots \vee P(x_n),$$

because this disjunction is true if and only if at least one of $P(x_1), P(x_2), \dots, P(x_n)$ is true.

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have **higher precedence than all logical operators** from propositional calculus.

For example, $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ and $Q(x)$.
In other words, it means

$$(\forall x P(x)) \vee Q(x) \text{ rather than } \forall x (P(x) \vee Q(x)).$$

Observe that the two statements “For all real number x , if x is an integer then x is rational”, and “For all integer x , x is rational” mean the same thing because the set of integers is a subset of the set of real numbers. Both have informal translations “All integers are rational”. In fact, a statement of the form

$$\forall x \in U [\text{if } P(x) \text{ then } Q(x)]$$

can always be rewritten in the form

$$\forall x \in D [Q(x)]$$

by narrowing U to be the subset D consisting of all values of the variable x that make $P(x)$ true. Conversely, a statement of the form

$$\forall x \in D [Q(x)]$$

can be rewritten as

$$\forall x [\text{if } x \in D \text{ then } Q(x)].$$

Implicit Quantification

Consider the statement

if a number is an integer, then it is a rational number

As shown earlier, this statement is equivalent to a universal statement. However, it does not contain the telltale word *all* or *every* or *any* or *each*. The only clue to indicate its universal quantification comes from the presence of the indefinite article “*a*”. This is an example of ***implicit*** universal quantification.

Existential quantification can also be ***implicit***. For instance, the statement “*The number 24 can be written as a sum of two even integers*” can be expressed formally as “ $\exists$ even integers m and n such that $24 = n + m$ ”.

Notation. Let $P(x)$ and $Q(x)$ be predicates and suppose the common domain of x is D .

- The notation $P(x) \Rightarrow Q(x)$ means that every element in the truth set of $P(x)$ is in the truth set of $Q(x)$, or, equivalently,

$$\forall x [P(x) \rightarrow Q(x)].$$

- The notation $P(x) \Rightarrow Q(x)$ means that $P(x)$ and $Q(x)$ have identical truth sets, or, equivalently,

$$\forall x [P(x) \leftrightarrow Q(x)].$$

Binding Variables

1. When a quantifier is used on the variable x , we say that this occurrence of the variable is **bound**.
2. An occurrence of a variable that is not bound by a quantifier or set equal to a particular value is said to be **free**.

All the variables that occur in a propositional function must be bound or set equal to a particular value to turn it into a proposition. This can be done using a combination of universal quantifiers, existential quantifiers, and value assignments.

Binding Variables

Example. In the statement

$$\exists x (x + y = 1),$$

- the variable x is bound by the existential quantification $\exists x$,
- but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable.

Negating Quantified Expressions

We will often want to consider the negation of a quantified expression. For instance, consider the negation of the statement

“Every student in your class has taken a course in calculus.”

This statement is a universal quantification, namely, $\forall xP(x)$, where $P(x)$ is the statement “ x has taken a course in calculus” and the domain consists of the students in your class.

The negation of this statement is “*It is not the case that every student in your class has taken a course in calculus.*” This is equivalent to “*There is a student in your class who has not taken a course in calculus.*” And this is simply the existential quantification of the negation of the original propositional function, namely,

$$\exists x \neg P(x).$$

Negations of Quantified Statements

Theorem. The negation of a statement of the form

$$\forall x \in D [Q(x)]$$

is logically equivalent to a statement of the form

$$\exists x \in D [\neg Q(x)]$$

Symbolically,

$$\neg (\forall x \in D [Q(x)]) \iff \exists x \in D [\neg Q(x)]$$

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Negating Quantified Expressions

TABLE 1 Quantifiers.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall xP(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists xP(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg\exists xP(x)$	$\forall x\neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall xP(x)$	$\exists x\neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

Negations of Quantified Statements

Example. What are the negations of the statements

a) $\forall x(x^2 > x)$

b) $\exists x(x^2 = 2)?$

Solution:

a) The negation of $\forall x(x^2 > x)$ is the statement $\neg\forall x(x^2 > x)$, which is equivalent to $\exists x\neg(x^2 > x)$. This can be rewritten as $\exists x(x^2 \leq x)$.

b) The negation of $\exists x(x^2 = 2)$ is the statement $\neg\exists x(x^2 = 2)$, which is equivalent to $\forall x\neg(x^2 = 2)$. This can be rewritten as $\forall x(x^2 \neq 2)$.
The truth values of these statements depend on the domain.

Example. Use predicates and quantifiers to express the system specifications “*Every mail message larger than one megabyte will be compressed*”

Solution:

Let $S(m, y)$ be “*Mail message m is larger than y megabytes,*” where the variable x has the domain of all mail messages and the variable y is a positive real number, and

let $C(m)$ denote “*Mail message m will be compressed.*”

Then the specification “*Every mail message larger than one megabyte will be compressed*” can be represented as

$$\forall m (S(m, 1) \rightarrow C(m))$$

Example. Use predicates and quantifiers to express the system specifications “*If a user is active, at least one network link will be available.*”

Solution:

Let $A(u)$ represent “*User u is active,*” where the variable u has the domain of all users;

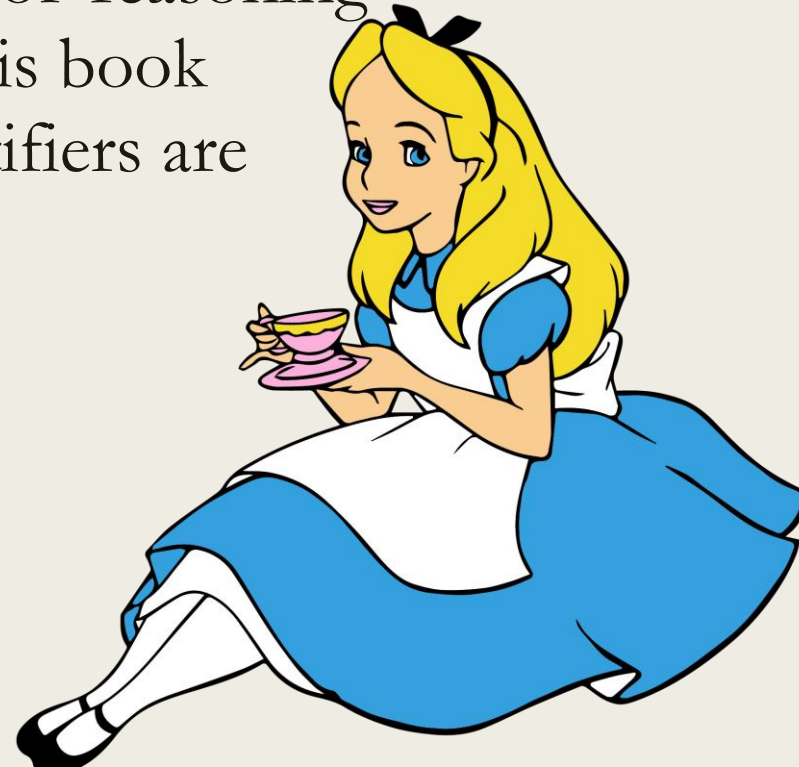
Let $S(n, x)$ denote “*Network link n is in state x ,*” where n has the domain of all network links and x has the domain of all possible states for a network link.

Then the specification “*If a user is active, at least one network link will be available*” can be represented by

$$\exists u A(u) \rightarrow \exists n S(n, \text{available}).$$

Examples from Lewis Carroll

Lewis Carroll (really C. L. Dodgson writing under a pseudonym), the author of *Alice in Wonderland*, is also the author of several works on symbolic logic. His books contain many examples of reasoning using quantifiers. The next examples come from his book *Symbolic Logic*; These examples illustrate how quantifiers are used to express various types of statements.



Examples from Lewis Carroll

Example. Consider these statements. The first two are called *premises* and the third is called the *conclusion*. The entire set is called an *argument*.

“All lions are fierce.”

“Some lions do not drink coffee.”

“Some fierce creatures do not drink coffee.”

Let $P(x)$, $Q(x)$, and $R(x)$ be the statements “ x is a lion,” “ x is fierce,” and “ x drinks coffee,” respectively. Assuming that the domain consists of all creatures, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, and $R(x)$.

Examples from Lewis Carroll

Solution: We can express these statements as

$$\forall x (P(x) \rightarrow Q(x)).$$

$$\exists x (P(x) \wedge \neg R(x)).$$

$$\exists x (Q(x) \wedge \neg R(x)).$$

Notice that the 2nd statement cannot be written as $\exists x (P(x) \rightarrow \neg R(x))$. The reason is that $P(x) \rightarrow \neg R(x)$ is true whenever x is not a lion, so that $\exists x (P(x) \rightarrow \neg R(x))$ is true as long as there is at least one creature that is not a lion, even if every lion drinks coffee. Similarly, the third statement cannot be written as

$$\exists x (Q(x) \rightarrow \neg R(x)).$$

Examples from Lewis Carroll

Example. Consider these statements, of which the first three are premises and the fourth is a valid conclusion.

“All hummingbirds are richly colored.”

“No large birds live on honey.”

“Birds that do not live on honey are dull in color.”

“Hummingbirds are small.”

Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements “ x is a hummingbird,” “ x is large,” “ x lives on honey,” and “ x is richly colored,” respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$, and $S(x)$.

Examples from Lewis Carroll

Soluiton: We can express the statements in the argument as

$$\begin{aligned}\forall x (P(x) \rightarrow S(x)). \\ \neg \exists x (Q(x) \wedge R(x)). \\ \forall x (\neg R(x) \rightarrow \neg S(x)). \\ \forall x (P(x) \rightarrow \neg Q(x)).\end{aligned}$$

Note we have assumed that “small” is the same as “not large” and that “dull in color” is the same as “not richly colored.”

Nested Quantifiers

If you want to establish the truth of a statement of the form

$$\forall x \in D \exists y \in E [P(x, y)]$$

your challenge is to allow someone else to pick whatever element x in D they wish and then you must find an element y in E that “works” for that particular x .

If you want to establish the truth of a statement of the form

$$\exists x \in D \forall y \in E [P(x, y)]$$

your goal is to find one particular x in D that will “work” no matter what y in E anyone might choose to challenge you with.

Translating from Informal to Formal Language

Example. “*There Is a Smallest Positive Integer*” is equivalent to the statement

$$\exists m \in \mathbb{Z}^+ \forall n \in \mathbb{Z}^+ [m \leq n]$$

Example. “*There Is No Smallest Positive real Number*” is equivalent to the statement

$$\forall x \in \mathbb{R}^+ \exists y \in \mathbb{R}^+ [y < x]$$

Example. The Definition of Limit of a Sequence:

$$\lim_{n \rightarrow \infty} a_n = L \Leftrightarrow \forall \varepsilon > 0 \exists n_0 \in \mathbb{N} \forall n \in \mathbb{N} [n > n_0 \Rightarrow |a_n - L| < \varepsilon]$$

Example. Translate into English the statement

$$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$$

where the domain for both variables consists of all real numbers.

Solution: This statement says that for every real number x and for every real number y , if $x > 0$ and $y < 0$, then $xy < 0$. That is, this statement says that for real numbers x and y , if x is positive and y is negative, then xy is negative. This can be stated more succinctly as

“The product of a positive real number and a negative real number is always a negative real number.”

Thinking of quantification as loops

In working with quantifications of more than one variable, it is sometimes helpful to think in terms of nested loops. (If there are infinitely many elements in the domain of some variable, we cannot actually loop through all values. Nevertheless, this way of thinking is helpful in understanding nested quantifiers.)

For example, to see whether $\forall x \forall y P(x, y)$ is true, we loop through the values for x , and for each x we loop through the values for y .

- If we find that for all values of x that $P(x, y)$ is true for all values of y , we have determined that $\forall x \forall y P(x, y)$ is true.
- If we ever hit a value x for which we hit a value y for which $P(x, y)$ is false, we have shown that $\forall x \forall y P(x, y)$ is false.

Thinking of quantification as loops

Similarly, to determine whether $\forall x \exists y P(x, y)$ is true, we loop through the values for x . For each x we loop through the values for y until we find a y for which $P(x, y)$ is true.

- If for every x we hit such a y , then $\forall x \exists y P(x, y)$ is true;
- if for some x we never hit such a y , then $\forall x \exists y P(x, y)$ is false.

Thinking of quantification as loops

To see whether $\exists x \forall y P(x, y)$ is true, we loop through the values for x until we find an x for which $P(x, y)$ is always true when we loop through all values for y .

- Once we find such an x , we know that $\exists x \forall y P(x, y)$ is true.
- If we never hit such an x , then we know that $\exists x \forall y P(x, y)$ is false.

Thinking of quantification as loops

Finally, to see whether $\exists x \exists y P(x, y)$ is true, we loop through the values for x , where for each x we loop through the values for y until we hit an x for which we hit a y for which $P(x, y)$ is true.

The statement $\exists x \exists y P(x, y)$ is false only if we never hit an x for which we hit a y such that $P(x, y)$ is true.

The order of quantifiers

NOTE: The order of nested universal quantifiers in a statement without other quantifiers can be changed without changing the meaning of the quantified statement.

Example. Let $P(x, y)$ be the statement “ $x + y = y + x$.” What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$, where the domain for all variables consists of all real numbers?

Solution: Both denotes the proposition “For all real numbers x , for all real numbers y , $x + y = y + x$.”

Answer: $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$, have the same meaning, and both are true.

NOTE: Be careful with the order of existential and universal quantifiers!

Example. Let $Q(x, y)$ denote “ $x + y = 0$.” What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

Solution: The quantification $\exists y \forall x Q(x, y)$ denotes the proposition “There is a real number y such that for every real number x , $Q(x, y)$.” No matter what value of y is chosen, there is only one value of x for which $x + y = 0$.

The quantification $\forall x \exists y Q(x, y)$ denotes the proposition “For every real number x there is a real number y such that $Q(x, y)$.” Given a real number x , there is a real number y such that $x + y = 0$; namely, $y = -x$. Hence, the statement $\forall x \exists y Q(x, y)$ is true.

The previous Example illustrates that the order in which quantifiers appear makes a difference. The statements $\exists y \forall x P(x, y)$ and $\forall x \exists y P(x, y)$ are not logically equivalent.

The statement $\exists y \forall x P(x, y)$ is true if and only if there is a y that makes $P(x, y)$ true for every x . So, for this statement to be true, there must be a particular value of y for which $P(x, y)$ is true regardless of the choice of x .

On the other hand, $\forall x \exists y P(x, y)$ is true if and only if for every value of x there is a value of y for which $P(x, y)$ is true. So, for this statement to be true, no matter which x you choose, there must be a value of y (possibly depending on the x you choose) for which $P(x, y)$ is true. In other words, in the second case, y can depend on x , whereas in the first case, y is a constant independent of x .

TABLE 1 Quantifications of Two Variables.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Example. Let $Q(x, y, z)$ be the statement “ $x + y = z$.” What are the truth values of the statements $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consists of all real numbers?

Solution: Suppose that x and y are assigned values. Then, there exists a real number z such that $x + y = z$. Consequently, the quantification $\forall x \forall y \exists z Q(x, y, z)$, which is the statement “*For all real numbers x and for all real numbers y there is a real number z such that $x + y = z$,*” is true.

The quantification $\exists z \forall x \forall y Q(x, y, z)$, which is the statement “*There is a real number z such that for all real numbers x and for all real numbers y it is true that $x + y = z$,*” is false, because there is no value of z that satisfies the equation $x + y = z$ for all values of x and y .

Practice

Task 1. Translate the statement

$$\forall x \left(C(x) \vee \exists y (C(y) \wedge F(x, y)) \right)$$

into English, where

- $C(x)$ is “ x has a computer,”
- $F(x, y)$ is “ x and y are friends,”
- and the domain for both x and y consists of all students in your school.

Task 1. Translate the statement

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into English, where

- $C(x)$ is “ x has a computer,”
- $F(x, y)$ is “ x and y are friends,”
- and the domain for both x and y consists of all students in your school.

Answer: The statement says that for every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends.

In other words, every student in your school has a computer or has a friend who has a computer.

Task 2. Translate the statement

$$\exists x \forall y \forall z \left((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z) \right)$$

into English, where

- $F(a, b)$ means a and b are friends and
- the domain for x , y , and z consists of all students in your school.

Task 2. Translate the statement

$$\exists x \forall y \forall z \left((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z) \right)$$

into English, where

- $F(a, b)$ means a and b are friends and
- the domain for x , y , and z consists of all students in your school.

Answer: There is a student x such that for all students y and all students z other than y , if x and y are friends and x and z are friends, then y and z are not friends.

In other words, there is a student none of whose friends are also friends with each other.

Task 3. Express the statement

“If a person is female and is a parent, then this person is someone’s mother”

as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Task 3. Express the statement “If a person is female and is a parent, then this person is someone’s mother” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The given statement can be expressed as “For every person x , if person x is female and person x is a parent, then there exists a person y such that person x is the mother of person y .” We introduce the propositional functions

- $F(x)$ to represent “ x is female,”
- $P(x)$ to represent “ x is a parent,” and
- $M(x, y)$ to represent “ x is the mother of y .”

The original statement can be represented as

$$\forall x \left((F(x) \wedge P(x)) \rightarrow \exists y M(x, y) \right).$$

We can move $\exists y$ to the left so that it appears just after $\forall x$, because y does not appear in $F(x) \wedge P(x)$. We obtain the logically equivalent expression

$$\forall x \exists y \left((F(x) \wedge P(x)) \rightarrow M(x, y) \right)$$

Task 4. Express the statement

“Everyone has exactly one best friend”

as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Task 4. Express the statement “*Everyone has exactly one best friend*” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The given statement can be expressed as “For every person x , person x has exactly one best friend.” This statement is the same as “ $\forall x$ (person x has exactly one best friend),” where the domain consists of all people.

To say that x has exactly one best friend means that there is a person y who is the best friend of x , and furthermore, that for every person z , if person z is not person y , then z is not the best friend of x . When we introduce the predicate

➤ $B(x, y)$ to be the statement “ y is the best friend of x ,”

The statement that “ x has exactly one best friend” can be represented as

$$\exists y \left(B(x, y) \wedge \forall z \left((z \neq y) \rightarrow \neg B(x, z) \right) \right).$$

Consequently, our original statement can be expressed as

$$\forall x \exists y \left(B(x, y) \wedge \forall z \left((z \neq y) \rightarrow \neg B(x, z) \right) \right).$$

Task 5. Use quantifiers to express the statement

“There is a woman who has taken a flight on every airline in the world.”

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Solution:

➤ Let $P(w, f)$ be “ w has taken f ” and

➤ $Q(f, a)$ be “ f is a flight on a .”

We can express the statement as

$$\exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$$

where the domains of discourse for w , f , and a consist of all the women in the world, all airplane flights, and all airlines, respectively.

Negating nested quantifiers

Example. Express the negation of the statement $\forall x \exists y(xy = 1)$ so that no negation precedes a quantifier.

Solution: By successively applying De Morgan's laws for quantifiers, we can move the negation in inside all the quantifiers. We find that

$\neg \forall x \exists y(xy = 1)$ is equivalent to $\exists x \neg \exists y(xy = 1)$, which is equivalent to $\exists x \forall y \neg(xy = 1)$.

We conclude that our negated statement can be expressed as

$$\exists x \forall y (xy \neq 1)$$

Negating nested quantifiers

Example. Use quantifiers to express the statement that “There does not exist a woman who has taken a flight on every airline in the world.”

Example. Use quantifiers to express the statement that “There does not exist a woman who has taken a flight on every airline in the world.”

Solution: $\neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$, where

➤ $P(w, f)$ is “ w has taken f ” and

➤ $Q(f, a)$ is “ f is a flight on a .”

$$\begin{aligned} & \forall w \neg \forall a \exists f (P(w, f) \wedge Q(f, a)) \equiv \\ & \equiv \forall w \exists a \neg \exists f (P(w, f) \wedge Q(f, a)) \equiv \\ & \equiv \forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a)) \equiv \\ & \equiv \forall w \exists a \forall f (\neg P(w, f) \vee \neg Q(f, a)). \end{aligned}$$

This last statement states “*For every woman there is an airline such that for all flights, this woman has not taken that flight or that flight is not on this airline.*”